SHoEbox: decoder cohort diversity

November 5, 2026 (UTC) | Twelve frozen graphs, 432 matched residual fits

More fitting data helps the linear corrections most. With the expanded cohort, validation retains 33 of 36 identity-activation corrections, up from 3. Three of the four graph conditions improve both private and shared accuracy. GELU gives smaller private-answer gains and still trades away shared accuracy in most conditions. The default decoder remains unchanged.

The main comparison expands correction training from 960 to 3,840 cases per graph. Graph weights, the original linear byte decoder and its normalization, and every validation/test/independent feature cache stay fixed. Both training cohorts are captured and fitted in the same runtime; the smaller cohort is an exact subset of the larger one.

Private answers on held-out fact combinations

Frozen graph	Frozen linear	960 cases	3,840 cases	Change
Original / dropout	73.7%	73.2%	77.3%	+4.1 pp
Original / balanced	56.0%	56.0%	58.0%	+2.0 pp
Gated / dropout	60.9%	60.9%	59.8%	-1.1 pp
Gated / balanced	54.9%	54.9%	56.6%	+1.7 pp

Shared answers on the same held-out combinations

Frozen graph	Frozen linear	960 cases	3,840 cases	Change
Original / dropout	87.0%	87.2%	88.1%	+0.9 pp
Original / balanced	80.5%	80.5%	82.8%	+2.3 pp
Gated / dropout	91.1%	91.1%	90.1%	-1.0 pp
Gated / balanced	87.5%	87.5%	89.0%	+1.5 pp

These tables show identity-activation corrections with validation-selected preservation penalty and scale. Private accuracy averages left and right queries. Each group averages three graph-training seeds after averaging three matched correction initializations per graph. Changes compare the two newly fitted cohort sizes and are measured in percentage points.

Original/dropout gains 4.1 percentage points on private answers and 0.9 on shared answers. Original/balanced gains 2.0 and 2.3 points; gated/balanced gains 1.7 and 1.5. Gated/dropout declines by 1.1 and 1.0 points.

Matched nonlinear corrections

Identity and GELU corrections have the same 16 hidden units, 5,392 trainable parameters, initial tensors and fitting settings. Both add to the original 16,640-parameter frozen linear head. The following controls also use validation-selected preservation penalty and scale.

Frozen graph	960 cases: private / shared	3,840 cases: private / shared
Original / dropout	74.0% / 86.4%	75.1% / 83.4%
Original / balanced	57.2% / 79.1%	59.0% / 78.0%
Gated / dropout	61.0% / 90.6%	62.3% / 90.7%
Gated / balanced	56.0% / 86.4%	56.6% / 86.3%

Ordinary loss, without the preservation penalty

These GELU corrections use the same cohorts, initializations and validation scale guards, with the training penalty fixed at zero.

Frozen graph	960 cases: private / shared	3,840 cases: private / shared
Original / dropout	73.6% / 87.2%	75.0% / 85.6%
Original / balanced	56.0% / 80.5%	60.0% / 78.5%
Gated / dropout	61.1% / 90.6%	62.9% / 90.6%
Gated / balanced	55.1% / 87.4%	55.6% / 87.6%

Validation choices and missing-source recovery

Correction family	Zero scale, 960 cases	Zero scale, 3,840 cases
Identity, preservation selected	33 / 36	3 / 36
GELU, preservation selected	18 / 36	14 / 36

Selected GELU private accuracy improves by 0.6 to 1.7 percentage points across the four conditions. Shared accuracy falls in three conditions and increases by 0.1 point on gated/dropout. Only two of the twelve individual graphs improve both private and shared accuracy with selected GELU. The ordinary-loss results also show that adding a preservation penalty does not consistently improve the larger-cohort GELU solution.

B-only shared answers

Frozen graph	Identity 960	Identity 3,840	GELU 960	GELU 3,840
Original / dropout	86.7%	90.4%	91.3%	90.1%
Original / balanced	100.0%	100.0%	100.0%	100.0%
Gated / dropout	95.8%	95.6%	96.8%	96.1%
Gated / balanced	98.7%	99.2%	99.0%	98.7%

Missing-information controls average 25.1% across both activations and cohort sizes with selected preservation, over all graphs; their four-answer prior is 25%. Every decoder still emits a choice among all 256 bytes.

What changed in the training data

Inventory, per frozen graph	Small cohort	Expanded cohort
Cases per supported source/query cell	64	256
Total training cases	960	3,840
Distinct episode identities	192	768
Distinct observed traces with identifiers removed	960	3,840
Source/query/fact/template signatures	869	2,360
Training fact combinations	32	32
Shared / left / right training cases	448 / 256 / 256	1,792 / 1,024 / 1,024

Unavailable bytes are masked before counting observed traces. Removing identifiers strips each source's seven-byte header. Fact/template signatures include the source/query cell, full fact tuple and templates of available sources. The added cases vary presentation within the existing training partition; held-out fact combinations are unchanged.

Validation contains 1,920 cases and 384 distinct identities. Test/all contains 384 cases; the independent controls contain 3,456. Training, validation, test and independent episode identities are disjoint. No query label or target byte is passed to a head's prediction method.

Fitting and selection protocol

Two cohort sizes, three penalty coefficients (0, 1, 10), two activations, three correction initializations and twelve graphs give 432 new fits. Each uses query-balanced cross entropy, L2 coefficient 0.001 and full-batch L-BFGS with a 400-iteration cap. 432 of 432 fits reach that cap. Maximum absolute final gradients range from 2.4e-06 to 0.022.

Positive preservation penalties protect shared training examples already answered correctly by the frozen base. The target margin is the smaller of the base's true-answer margin and 1. The extra loss is the squared shortfall from that target. Targets and anchors come from training only.

Validation chooses a scale from 0, 0.125, 0.25, 0.5 and 1, minimizing query-balanced cross entropy while retaining at least the base's exact-correct counts on all-source/shared and B-only/shared validation cases. It then chooses between positive penalties 1 and 10. Ties prefer the smaller scale, then the smaller penalty. Selection locks are written before test scoring.

The cohorts share the same iteration cap, not the same arithmetic work: the larger arm processes four times as many cases per iteration. Recorded main-study capture executes 96,768,000 experts across twelve graphs, at K=1 and 84 expert executions per external event. Fitting and selection use cached features and make no graph-expert calls.

Runtime restoration and verification

Recapturing the original cases after runtime restoration was not bitwise identical to the old cache. Per-graph maximum absolute raw-feature differences range from 1.9e-06 to 0.00039. Both comparison arms therefore use fresh captures and fresh fits in one runtime. Legacy heads remain separate references in the aggregates. The loaded base weights and normalization stay exactly fixed; their independent numerical reconstruction uses 1e-12 absolute and relative tolerances.

All **305 automated tests pass**, including ten new cohort checks. Those ten were repeated after the normalization portability fix. The live output-slot example passes. Artifact audits reconstruct every input cohort, all 432 training objectives, validation choices, selected tensors, predictions and evaluation losses. Original cached evaluations and base predictions reproduce. The complete package includes every prerequisite checkpoint, feature cache, correction and prediction row.

Reproduce and continue

```
python scripts/run_diversity_study.py \
  --out artifacts/diversity_repeat --jobs 4 --threads 2
python -m experiments.text_regions.diversity_compare \
  --source artifacts/cohort_diversity_v1 \
  --out artifacts/reproduced_diversity_comparison.json
python scripts/check_all.py
```

Next refit the single full-rank linear byte decoder on the expanded cohort, then compare it with these identity and GELU corrections on the same fixed validation and evaluation data. This tests whether the broader training coverage can deliver the useful gains through one emission path, removing the additional correction branch. Measure inference cost alongside accuracy before changing the default.